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RESEARCH INTERESTS

AI for energy systems; Agentic AI and multimodal remote sensing; trustworthy and explainable machine learning; industrial process monitoring and fault diagnosis; robustness, formal verification, and reliable decision-making.

PROFILE

Schmidt AI in Science Postdoctoral Fellow at Cornell University developing trustworthy and agentic AI for energy, engineering, and scientific systems. My current research combines remote sensing, foundation models, and specialized analytical tools to map rooftop solar PV and analyze adoption and deployment across U.S. and international cities. My recent *Applied Energy* paper examines household and large-roof PV adoption pathways. This work builds on research in explainability, robustness, formal verification, and industrial fault diagnosis to support interpretable and reliable analysis of energy and industrial systems.

EDUCATION

Zhejiang University, Hangzhou, China 2018–2023
Ph.D. in Control Science and Engineering
Thesis: *Research on the Reliability of Data-driven Fault Diagnosis Models in Industrial Systems*

Zhejiang University of Technology, Hangzhou, China 2014–2018
B.Eng. in Automation and Electrical Engineering

EXPERIENCE

Cornell University, Ithaca, NY, USA
Schmidt AI in Science Postdoctoral Fellow, College of Engineering 2025–present

- Develop AI and remote-sensing workflows for rooftop PV mapping and adoption analysis across major U.S. and global cities, including a 15-city U.S. study and international border-city comparisons.
- Lead agentic vision–language workflows that coordinate foundation models and specialized tools for city-scale aerial-image analysis of rooftop-solar opportunity and deployment; analyze household and large-roof adoption pathways in *Applied Energy*.
- Develop trustworthy scientific AI with failure analysis, uncertainty-aware evaluation, and domain-scientist feedback, alongside interpretable pipelines for 3D plant phenotyping and maize reconstruction.

Nanyang Technological University, Singapore
Postdoctoral Research Fellow, School of Electrical and Electronic Engineering 2024

- Studied robustness, failure modes, and reliable decision-making in multi-agent reinforcement learning under perturbations and environment shifts.
- Collaborated with Continental on multi-robot planning algorithms for warehouse operations.

Zhejiang University, Hangzhou, China
Ph.D. Researcher, Control Science and Engineering 2018–2023

- Collaborated with an ammonia fertilizer plant and steel manufacturers on data-driven process monitoring and fault diagnosis, including reliable modeling under changing operating conditions.
- Developed Shapley-value-based IG^2 and ABIGX for AI feature attribution and unified explainable fault diagnosis.
- Developed MILP-based formal verification and optimization-based defenses for data-driven fault diagnosis models under adversarial perturbations, data scarcity, and distribution shift.
- Applied Bayesian optimization to safety–accuracy trade-offs and developed data augmentation methods for scarce and imbalanced industrial fault data.

FELLOWSHIPS & FUNDING

Schmidt AI in Science Postdoctoral Fellowship, Cornell University 2025–present

PUBLICATIONS

1. **Y. Zhuo** and F. You, "AI-enabled remote sensing reveals two urban rooftop PV adoption pathways: household vs. large-roof," *Applied Energy*, vol. 426, Art. no. 128766, 2026.
2. **Y. Zhuo** and Z. Ge, "IG2: Integrated Gradient on Iterative Gradient Path for Feature Attribution," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024.
3. **Y. Zhuo**, J. Qian, J. Zheng, X. Jiang, Z. Song, D. Chen, W. Yu, and Z. Ge, "ABIGX: A unified framework for eXplainable fault detection and classification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026.
4. **Y. Zhuo**, Z. Song, and Z. Ge, "Security versus accuracy: Trade-off data modeling to safe fault classification systems," *IEEE Transactions on Neural Networks and Learning Systems*, 2023.

5. **Y. Zhuo** and Z. Ge, "Adversarial security verification of data-driven FDC systems," *IEEE Transactions on Reliability*, 2022.
6. **Y. Zhuo** and Z. Ge, "Auxiliary information-guided industrial data augmentation for any-shot fault learning and diagnosis," *IEEE Transactions on Industrial Informatics*, 2021.
7. **Y. Zhuo** and Z. Ge, "Attack and defense: Adversarial security of data-driven FDC systems," *IEEE Transactions on Industrial Informatics*, 2022.
8. **Y. Zhuo**, YAW Shardt, and Z. Ge, "One-variable attack on the industrial fault classification system and its defense," *Engineering*, 2022.
9. **Y. Zhuo** and Z. Ge, "Data guardian: A data protection scheme for industrial monitoring systems," *IEEE Transactions on Industrial Informatics*, 2021.
10. J. Zheng, **Y. Zhuo**, et al., "Advances in Bayesian networks for industrial process analytics: Bridging data and mechanisms," *Expert Systems with Applications*, 2025.
11. **Y. Zhuo** and F. You, "3D plant phenotyping from a single image: learning fine-scale organ morphology with monocular depth estimation," *Computers and Electronics in Agriculture*, 2025.
12. **Y. Zhuo** and F. You, "MIRAGE: Biomechanically interpretable 3D generation and reconstruction of maize plants," *Computers and Electronics in Agriculture*, 2026.
13. **Y. Zhuo** and Z. Ge, "Gaussian discriminative analysis aided GAN for imbalanced big data augmentation and fault classification," *Journal of Process Control*, 2020.
14. J. Zheng, J. Wang, L. Ye, and **Y. Zhuo**, "Semi-supervised convolutional generative adversarial networks for dynamic fault classification with manifold regularization," *Process Safety and Environmental Protection*, 2025.
15. X. Jiang, C. Zheng, **Y. Zhuo**, et al., "Advancing Industrial Data Augmentation in AIGC Era: from Foundations to Frontier Applications," *IEEE Transactions on Instrumentation and Measurement*, 2025.
16. X. Zhang, B. Chen, **Y. Zhuo**, Z. Song, and X. Zhang, "Contrastive-based bi-update class incremental model and its application in fault diagnosis," *IEEE Sensors Journal*, 2024.
17. Z. Yin, **Y. Zhuo**, and Z. Ge, "Transfer adversarial attacks across industrial intelligent systems," *Reliability Engineering & System Safety*, 2023.

BOOK CHAPTER

1. Z. Ge, **Y. Zhuo**, and Z. Song, "On Security of Data-Driven FDC Systems: Attack, Defense, and Verification," in *Process Monitoring, Fault Diagnosis, and Tolerant Control for Complex Industrial Systems: A Festschrift in Honor of Professor Steven X. Ding on the Occasion of His Retirement*. Singapore: Springer Nature Singapore, 2025, pp. 145–191.

MANUSCRIPTS

1. **Y. Zhuo** and Z. Ge, "Feature augmentation for adversarial robustness," submitted to *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026.
2. **Y. Zhuo** and F. You, "Border cities reveal how rooftop solar opportunity becomes urban deployment," under review at *Nature Communications*, 2026. [GitHub]
3. **Y. Zhuo** and F. You, "Stock and flow structure of rooftop solar across US metropolitan areas," submitted to *Nature Cities*, 2026. [GitHub]

TEACHING

Guest Lecturer, SYSEN 6880: Deep Learning, Cornell University Two semesters
Delivered guest lectures on *Explainable AI* (feature attribution, Shapley-based explanations, counterfactuals) and *Advancing Computer Vision* (vision transformers, semantic segmentation, 3D reconstruction).

PROFESSIONAL SERVICE

Peer Reviewer for Journals: *IEEE Transactions on Industrial Informatics*; *IEEE Transactions on Automatic Control*; *IEEE Transactions on Systems, Man, and Cybernetics: Systems*; *Scientific Reports*; *IFAC Journal of Systems and Control*; *Pattern Recognition*.